

UNIT 9

CAPACITORS



CAPACITOR:**Definition:**

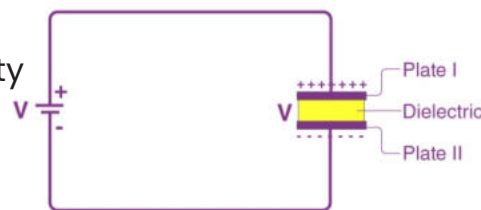
"The system of two isolated conductors that has capacity for storing electrical energy and electrical charge is called capacitor."

Working principle:

Typical capacitor consists of two parallel plates placed on insulating stand and are separated by some insulating material known as dielectric. The charge stored in the capacitor is directly proportional to the potential difference V across the plates of the capacitor.

$$q \propto CV$$

$$q = CV$$



Where 'C' is constant called capacitance of capacitor.

CAPACITANCE OF CAPACITOR:

"The charge stored per unit potential difference between the plates of capacitor is called capacitance of capacitor."

$$\text{Capacitance} = \frac{\text{Charge}}{\text{Potential difference}}$$

$$C = \frac{q}{V}$$

Unit:

S.I. unit of capacitance is Coulomb/Volt (C/V) or Farad.

TYPES OF CAPACITORS:

Capacitors play a very important role in various electrical circuits, especially in radio circuits. Here we classify some common types of capacitors:

1. Variable Capacitors:

This type of capacitor is used to control the capacitance continuously for tuning transmitters and receivers' signals.

2. Film Capacitors:

Film capacitors consisting of a relatively large family of capacitors with the difference being in their dielectric properties. They are available in almost any value and voltages as high as 1500 volts. Film capacitors are used in power electronics devices, phase shifters, X-ray flashes, and pulsed lasers.

3. Ceramic Capacitors:

These capacitors are used in high-frequency circuits ranging from audio to radio frequency. These capacitors are also called disc capacitors.

4. Electrolytic Capacitors: These capacitors are generally used when very large capacitance values are required. The majority of electrolytic capacitors are polarized. Electrolytic Capacitors are generally used in DC power supply circuits due to their large capacitances and small size, which help to reduce the ripple voltage.

CAPACITANCE OF PARALLEL PLATE CAPACITOR (AIR AS DIELECTRIC)

The electric intensity between two oppositely charged plates from Gauss's law is:

$$E = \frac{\sigma}{\epsilon_0} \dots\dots 1$$

Where σ is the surface charge density of the plate, which is the total charge per unit area.

$$\sigma = \frac{q}{A} \dots\dots\dots 2$$

Thus, equation (1) becomes:

$$E = \frac{q}{\frac{A}{\epsilon_0}}$$

$$E = \frac{q}{\epsilon_0 A} \dots\dots\dots 3$$

Where,

$$E = \frac{V}{d} \dots\dots\dots 4$$

By comparing equation (3) and (4),

$$\frac{V}{d} = \frac{q}{\epsilon_0 A}$$

$$V = \frac{qd}{\epsilon_0 A} \dots\dots\dots 5$$

As we know that, capacitance of any capacitor is,

$$C = \frac{q}{V} \dots\dots\dots 6$$

By substituting the value of V in equation (6), from (5) we get,

$$C = \frac{q}{\frac{qd}{\epsilon_0 A}}$$

$$C = \frac{\epsilon_0 Aq}{qd}$$

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance of Parallel Plate Capacitor (Dielectric Slab as Medium):

If an insulating material or dielectric of relative permittivity ϵ_r is placed between the plates of the capacitor, then the capacitance of the capacitor is given by:

$$C' = \frac{\epsilon A}{d}$$

But $\epsilon = \epsilon_0 \epsilon_r$ therefore:

$$C' = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C' = \epsilon_r \frac{\epsilon_0 A}{d}$$

$$C' = \epsilon_r C$$

Thus, the above result shows that the capacitance of the capacitor increases in the presence of a dielectric other than air.

COMBINATIONS OF PARALLEL PLATE CAPACITORS:

1. Parallel Combination:

Definition:

"Capacitors are said to be connected in parallel when they are joined across the same two points."

Expression for Equivalent Capacitance:

Consider three capacitors C_1 , C_2 , and C_3 in parallel. In this connection, the same potential difference P.d. exists across each capacitor, but different charges are stored by each capacitor. Suppose the applied potential difference is V .

Now,

- Charge on capacitor C_1 is $Q_1 = C_1 V$... (1)
- Charge on capacitor C_2 is $Q_2 = C_2 V$... (2)
- Charge on capacitor C_3 is $Q_3 = C_3 V$... (3)

Consider an equivalent capacitor of capacitance C_e .

If the applied potential is V , and total charge stored is Q , then:

$$q = C_e V$$

Now,

$$Q = Q_1 + Q_2 + Q_3$$

$$C_e V = C_1 V + C_2 V + C_3 V$$

$$C_e V = V(C_1 + C_2 + C_3)$$

$$C_e = C_1 + C_2 + C_3$$

For 'n' numbers of capacitors connected in series,

$$C_e = C_1 + C_2 + C_3 + \dots + C_n$$

- The equivalent capacitance is greater than the greatest individual capacitance.
- Capacitor in parallel may be regarded as a single capacitor of greater plate area.
- For 'n' equal capacitors in parallel, equivalent capacitance $C_e = nC$.

2. Series Combination:

Definition:

"Capacitors are said to be connected in series when they are joined end to end."

Expression for Equivalent Capacitance:

Consider three capacitors of capacitances C_1 , C_2 , and C_3 in series. In this connection, the same charge is stored by each capacitor, but different potential differences (P.d.) exist across each capacitor.

Suppose, applied P.d. = V .

Now:

- P.d. across the capacitor C_1 is $V_1 = Q/C_1$ 1
- P.d. across the capacitor C_2 is $V_2 = Q/C_2$ 2
- P.d. across the capacitor C_3 is $V_3 = Q/C_3$ 3

Consider an equivalent capacitor of capacitance C_e . If the applied voltage is V , then:

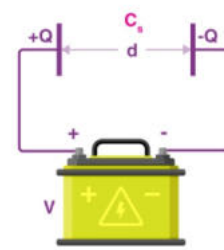
$$V = \frac{Q}{C_e} \dots \dots \dots 4$$

$$V = V_1 + V_2 + V_3 \dots \dots \dots 5$$

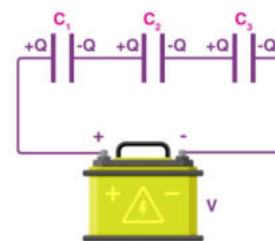
$$\frac{Q}{C_e} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

$$\frac{Q}{C_e} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

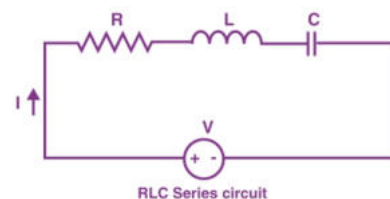
$$\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



(b)



(a)



RLC Series circuit

For 'n' number of capacitors:

$$\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

Characteristics of Series Combinations:

- The equivalent capacitance is less than the smallest individual capacitance.
- Capacitor in series may be regarded as a single capacitor of greater plate separation.
- For 'n' equal capacitors in series, equivalent capacitance $C_e = C/n$.

CHARGING AND DISCHARGING OF CAPACITOR:

Consider a circuit having a capacitance C and a resistance R , which are connected in series with a battery of potential difference V through a switch as shown in the figure. This circuit is called an RC circuit.

Charging of Capacitor:

When the switch is at position 1 as shown in Fig. (a), the capacitor begins to store charge. If at any time during charging, the charge Q flowing through the circuit and Q is the charge on the capacitor, then the sum of the potential difference between the plates of the capacitor and across the resistor is equal to the potential difference supplied by the battery. As the current stops flowing when the capacitor is fully charged, when $Q = Q_0$ (the maximum value of the charge on the capacitor), then:

$$Q_0 = CV$$

Experiment shows that the charging process of the capacitor exhibits exponential behavior.

Therefore, we can write its equation as:

$$Q = Q_0(1 - e^{-t})$$

$$Q = Q_0(1 - \frac{1}{e})$$

$$Q = Q_0(1 - \frac{1}{2.718})$$

$$Q = Q_0(1 - 0.368)$$

$$Q = 0.632Q_0$$

Therefore, the time constant is the duration of time for the capacitor in which 63.2% of its maximum charge is deposited on the plates during the charging of the capacitor.